

Multi-Actor DRL with LLM for Energy-Efficient Sparse Hybrid RIS-IAB Network

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Abstract—Integrated access and backhaul (IAB) networks for large-scale Internet of Things (IoT) applications face critical challenges in achieving high energy efficiency (EE) and cost-effective deployment due to the dense use of active transceivers and increased hardware complexity. Reconfigurable intelligent surfaces (RIS) can enhance coverage but suffer from double fading and limited flexibility. This study investigates a sparse Hybrid RIS (HRIS)-IAB architecture that combines active and passive RIS elements while activating only a subset of RIS to balance performance and energy cost. We formulate an EE maximization problem to jointly optimize the sparse HRIS control and active element selection under access-backhaul coupling and hardware constraints. We developed a large language model (LLM)-multilayer perceptron (MLP) multi-actor deep reinforcement learning (DRL) framework built on a deep deterministic policy gradient, combining semantic reasoning with numerical policy learning. Numerical results show that for an increasing number of RIS elements, the proposed method achieved energy efficiencies that are 17.4% and 42.1% higher than those of the LLM-only single-actor and MLP-only single-actor DRL algorithms, respectively.

Index Terms—Deep reinforcement learning, Internet-of-Things, Integrated access and backhaul, large language models, and reconfigurable intelligent surfaces.

I. INTRODUCTION

The growth of the Internet of Things (IoT) is a defining force in the evolution of fifth- and sixth-generation (5G/6G) networks. With tens of billions of connected devices spanning smart cities, industrial automation, healthcare, and intelligent transportation systems, next-generation infrastructures must ensure reliable coverage in obstructed environments while scaling with device density and minimizing power consumption to meet sustainability goals. Integrated Access and Backhaul (IAB) has emerged as a key paradigm by reusing spectrum for both access and backhaul links, enabling flexible multi-hop deployments in dense millimeter-wave (mmWave) networks [1]. IAB reduces fiber dependence and supports resilient connectivity for mobile users, yet its energy efficiency (EE) is affected by severe path loss and blockages at high frequencies. These limitations can be mitigated through reconfigurable intelligent surface (RIS)-assisted signal control. RISs dynamically reshape wireless propagation via tunable reflection coefficients [2], enhancing coverage and spectral efficiency without adding active radio frequency (RF) chains. However, fully passive RISs suffer from multiplicative fading in cascaded channels, causing signal attenuation in long-range or high-mobility scenarios [3],

while active RISs counter this with amplifiers at the cost of higher circuit power, noise, and cost. Sparse HRIS schemes further activate only selected elements [4], achieving a trade-off between performance, hardware complexity, and power consumption, making HRIS particularly suited for scalable, energy-aware IAB systems. To fully exploit hybrid RIS gains in multi-hop IAB networks, delay alignment modulation (DAM) synchronizes precoding across direct and RIS-assisted paths, mitigating inter-symbol interference and enabling coherent combining, which is critical for maximizing rate gains when access and backhaul links operate jointly. [5]

Several works have reviewed the foundations of RIS and IAB systems, covering hardware architectures, deployment, resource allocation, and EE challenges for beyond-5G networks. HRIS has been proposed to mitigate the multiplicative fading effect inherent to fully passive RISs [6]. Subsequent studies have optimized HRIS designs for objectives such as rate maximization [7], user fairness [8], physical layer security (PLS) [9], and EE [10]. However, these works often rely on convex or iterative optimization under static and perfectly known channels, limiting their scalability in dynamic IoT environments. Deep reinforcement learning (DRL)-based methods such as multi-agent deep deterministic policy gradient (DDPG) have shown promise [3], yet primarily target sum-rate optimization of RIS-aided networks rather than EE-aware HRIS control for IAB networks. Sparse or on/off RIS schemes activating only selected elements have been explored to reduce circuit power. [9], though most adopt heuristic selection without adaptive intelligence. While RIS-IAB integration has been studied for interference mitigation and joint beamforming [11], the incorporation of DAM into RIS-IAB architectures remains unaddressed.

Existing research leaves several open challenges: First, RIS sparsification methods lack adaptive and learning-based control for energy-aware operation. Second, HRIS and IAB designs are typically optimized independently, neglecting access-backhaul coupling and realistic hardware power constraints. Third, RIS-IAB and DAM studies are treated separately, leaving end-to-end EE unexplored. We propose an LLM-driven multi-actor (MA) DRL framework for sparse HRIS-assisted IAB networks, enabling joint element selection, beamforming, and control optimization to enhance overall EE. The main contributions of this study are summarized as follows:

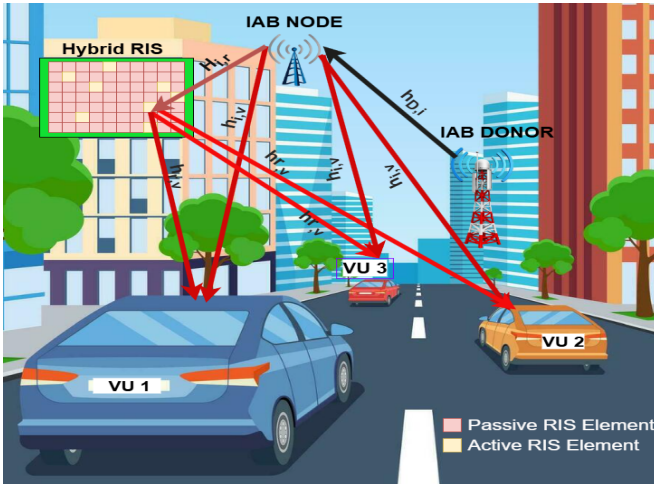


Fig. 1: An illustration of HRIS-aided IAB network.

- We propose an energy-efficient sparse HRIS-assisted IAB architecture integrating active-passive element configuration, DAM-based transmission, and realistic hardware power models for dense IoT deployments.
- We formulate an EE maximization problem to jointly optimize IAB beamforming, HRIS phase shifts, amplification coefficients, active element selection, and power allocation under access-backhaul coupling and QoS constraints.
- We develop a multi-agent DRL (MA-DRL) based DDPG algorithm with an MLP+LLM actor that combines numerical precision with semantic reasoning for high-dimensional continuous learning.
- Numerical results demonstrate that the proposed MA model achieves 17.4% higher EE than the LLM-only actor and 42.1% higher than the MLP baseline, confirming its effectiveness for HRIS-IAB optimization.

II. SYSTEM MODEL

We consider an HRIS-assisted IAB system designed for IoT-enabled vehicular users (VUs), illustrated in Fig. 1. The network consists of V single-antenna VUs, indexed by $\mathcal{V} = \{1, \dots, V\}$, and I IAB nodes, indexed by $\mathcal{I} = \{1, \dots, I\}$. Each IAB node aggregates VU data and forwards it to a central IAB donor via wireless backhaul links. An in-band backhaul scheme is assumed. The total bandwidth B is divided by a splitting ratio $\beta \in (0, 1)$, allocating βB to access and $(1 - \beta)B$ to backhaul transmissions. Each IAB node has M antennas and is assisted by R HRISs, each containing N_r reflecting elements indexed by $\mathcal{N}_r = \{1, \dots, N_r\}$.

The operating mode of element $n \in \mathcal{N}_r$ is denoted by $s_{r,n} \in \{0, 1, 2\}$, representing passive, active, or off states, enabling hybrid active-passive surface control.

A. HRIS Reflection Model

Let $s_{r,n} \in \{0, 1, 2\}$ denote the operational mode of element n at the r th RIS. The complex reflection coefficient is

$$\phi_{r,n} = \mu_{r,n} e^{j\omega_{r,n}}, \quad \omega_{r,n} \in [0, 2\pi), \quad \forall n \in \mathcal{N}_r, \quad \forall r \in \mathcal{R}, \quad (1)$$

where $\omega_{r,n}$ is the induced phase shift and $\mu_{r,n}$ is the reflection (or amplification) magnitude. The value of $\mu_{r,n}$ depends on the mode:

$$\mu_{r,n} = \begin{cases} \in [1, \mu_{\max}], & \text{if } s_{r,n} = 2 \text{ (Active),} \\ 1, & \text{if } s_{r,n} = 1 \text{ (Passive),} \\ 0, & \text{if } s_{r,n} = 0 \text{ (Off).} \end{cases} \quad (2)$$

The r th HRIS reflection matrix is then given by

$$\Theta_r = \text{diag}(\phi_{r,1}, \dots, \phi_{r,N_r}) \in \mathbb{C}^{N_r \times N_r}. \quad (3)$$

The total number of RIS elements is distributed among three operational modes, i.e., Off, Passive, and Active, as follows:

$$\sum_{n=1}^N \mathbb{I}(s_{r,n} = 0) + \sum_{n=1}^N \mathbb{I}(s_{r,n} = 1) + \sum_{n=1}^N \mathbb{I}(s_{r,n} = 2) = N, \quad (4)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

A fixed number N_o of RIS elements are assigned to the Off mode to ensure EE and enforce sparsity.

$$\sum_{n=1}^N \mathbb{I}(s_{r,n} = 0) = N_o. \quad (5)$$

The remaining $N - N_o$ elements are adaptively divided between Active and Passive modes, satisfying

$$\sum_{n=1}^N \mathbb{I}(s_{r,n} = 1) + \sum_{n=1}^N \mathbb{I}(s_{r,n} = 2) = N - N_o. \quad (6)$$

These constraints allow dynamic mode selection for Active and Passive elements while maintaining a fixed power-saving Off-mode baseline.

B. Path-loss Model of Wireless Links

The path-loss model for a wireless communication link between a transmitter t (TX) and a receiver q (RX), accounting for both line-of-sight (LoS) and non-line-of-sight (NLoS) conditions, is expressed as

$$PL_{t,q} = \rho_0 \Xi_{t,q} d_{t,q}^\alpha, \quad \forall PL_{t,q} \in \{PL_{i,v}, PL_{r,v}, PL_{D,i}\}, \quad (7)$$

where $d_{t,q}$ is the distance between TX t and RX q , ρ_0 is the reference path loss, and $\alpha > 0$ is the path-loss exponent. The variable $\Xi_{t,q} \in \{\xi^{\text{LoS}}, \xi^{\text{NLoS}}\}$ is a Bernoulli random variable dependent on $d_{t,q}$, where ξ^{LoS} and ξ^{NLoS} are the attenuation constants for LoS and NLoS conditions, respectively. The reference loss is given by $\rho_0 \triangleq (\frac{4\pi f_c}{c})^2$, where f_c is the carrier frequency and c is the speed of light, with the condition $1 < \xi^{\text{LoS}} < \xi^{\text{NLoS}}$.

Furthermore, the probability of the event $\{\Xi_{t,q} = \xi^{\text{LoS}}\}$ for a given distance $d_{t,q}$, denoted by $\chi_{t,q}^{\text{LoS}} \triangleq \mathbb{P}[\Xi_{t,q} = \xi^{\text{LoS}} \mid d_{t,q}]$, can be characterized as $\chi_{t,q}^{\text{LoS}} = \left(1 + \varphi e^{-\alpha \left(\frac{180}{\pi} \sin^{-1}(d_{t,q}) - \varphi\right)}\right)^{-1}$, where φ and α are positive environment-specific parameters determined by the operating frequency and the propagation conditions. The probability of NLoS transmission is $\chi_{t,q}^{\text{NLoS}} = 1 - \chi_{t,q}^{\text{LoS}}$. The average path loss between TX t and RX r is

$$\overline{PL}_{t,q} = \rho_0 (\chi_{t,q}^{\text{LoS}} \xi^{\text{LoS}} + \chi_{t,q}^{\text{NLoS}} \xi^{\text{NLoS}}) d_{t,q}^\alpha. \quad (8)$$

C. Channel Model

The channel vector from the HRIS r to VU v , $\mathbf{h}_{r,v} \in \mathbb{C}^{N_r \times 1}$, is expressed as

$$\mathbf{h}_{r,v} = \sqrt{\frac{1}{PL_{r,v}}} [h_{r,v}^0, h_{r,v}^1, \dots, h_{r,v}^{N_r-1}]^T, \quad (9)$$

where $h_{r,v}^n \triangleq e^{-j\frac{2\pi}{\lambda}nd_{r,v}\phi_{r,v}}$ for $n \in \{0, \dots, N_r - 1\}$, λ is the carrier wavelength, $d_{r,v}$ is the normalized inter-element spacing, and $\phi_{r,v}$ denotes the cosine of the angle of arrival (AoA) of the signal at the HRIS from VU v . A similar channel model is opted for the IAB node to VU v denoted by $\mathbf{h}_{i,v} \in \mathbb{C}^{M \times 1}$, this following Rician fading model is followed by the channel from IAB node i to the r HRIS:

$$\mathbf{H}_{i,r} = \sqrt{\frac{1}{PL_{i,r}}} \left(\sqrt{\frac{\kappa}{1+\kappa}} \mathbf{H}_{i,r}^{\text{LoS}} + \sqrt{\frac{1}{1+\kappa}} \mathbf{H}_{i,r}^{\text{NLoS}} \right), \quad (10)$$

where $\mathbf{H}_{i,r} \in \mathbb{C}^{N_r \times M}$, κ is the Rician factor, $\mathbf{H}_{i,r}^{\text{LoS}}$ models the deterministic LoS component, and $\mathbf{H}_{i,r}^{\text{NLoS}} \sim \mathcal{CN}(0, \mathbf{I}_{N_r \times M})$ models the scattered component. The LoS component is given by

$$\mathbf{H}_{i,r}^{\text{LoS}} = [\mathbf{a}_{\text{HRIS}}(\phi_{i,r}) \cdot \mathbf{a}_{\text{IAB}}^H(\phi_i)], \quad (11)$$

where $\mathbf{a}_{\text{HRIS}}(\phi_{i,r})$ and $\mathbf{a}_{\text{IAB}}(\phi_i)$ are array response vectors at HRIS r and IAB node i , respectively.

The backhaul channel from IAB-donor D to IAB node i is modeled as

$$\mathbf{h}_{D,i} = \sqrt{\frac{1}{PL_{D,i}}} \tilde{\mathbf{h}}, \quad (12)$$

where $\tilde{\mathbf{h}} \sim \mathcal{CN}(0, \mathbf{I}_M)$ if the donor has M antennas. Given that both the donor and IAB nodes are typically deployed at high altitudes, the backhaul link is dominated by strong LoS components.

D. Signal model

Let $\mathbf{x}[k|\tau] \in \mathbb{C}^M$ denote the downlink signal vector transmitted from the IAB node to a VU at time slot k with delay τ . Following the DAM scheme, $\mathbf{x}[k|\tau]$ is given by

$$\mathbf{x}[k|\tau] = \mathbf{W}_1 \mathbf{s}[k - \tau] + \mathbf{W}_2 \mathbf{s}[k], \quad (13)$$

where $\mathbf{s}[k] \in \mathbb{C}^V$ is the baseband symbol vector at time k , and $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{C}^{M \times V}$ are the precoding matrices corresponding to the symbols with and without delay, respectively.

Considering both the direct IAB-VU link and the HRIS-assisted link, the received signal at VU v is expressed as

$$y_v[k] \triangleq \underbrace{\mathbf{h}_{i,v}^H \mathbf{x}[k|\tau] \delta[k - k_1]}_{\text{Direct link}} + \underbrace{\mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{H}_{i,R} \mathbf{x}[k|\tau] \delta[k - k_2]}_{\text{HRIS-assisted link}} + \underbrace{\mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{n}_a}_{\text{Amplifier noise}} + z_v[k], \quad (14)$$

where $\mathbf{h}_{i,v} \in \mathbb{C}^{M \times 1}$ is the direct IAB-VU channel, $\mathbf{H}_{i,R} \in \mathbb{C}^{N_r \times M}$ is the IAB-HRIS channel, $\mathbf{h}_{R,v} \in \mathbb{C}^{N_r \times 1}$ is the HRIS-VU channel, $\mathbf{n}_a \sim \mathcal{CN}(0, \sigma_a^2 \mathbf{D}_a)$ is the amplifier noise vector injected only by the active RIS elements with variance σ_a^2 , and $\mathbf{D}_a = \text{diag}(d_1, \dots, d_{N_r})$ with

$$d_n = \begin{cases} 1, & \text{if element } n \text{ is active,} \\ 0, & \text{if element } n \text{ is passive,} \end{cases}$$

$z_v[k] \sim \mathcal{CN}(0, \sigma_v^2)$ is thermal noise, and k_1 and k_2 denote the delays of the direct and HRIS-assisted paths, respectively.

Zero-forcing (ZF) constraints are applied to suppress inter-symbol interference (ISI):

$$\mathbf{h}_{i,v}^H \mathbf{W}_2 = 0, \quad \mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{H}_{i,R} \mathbf{W}_1 = 0, \quad (15)$$

where $\mathbf{h}_{i,v}$ is the direct IAB-VU channel and $\mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{H}_{i,R}$ represents the effective cascaded HRIS channel.

Under the ZF condition, the instantaneous SNR at VU v becomes

$$\Gamma_v \triangleq \frac{|\mathbf{h}_{i,v}^H \mathbf{W}_1 + \mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{H}_{i,R} \mathbf{W}_2|^2}{\sigma_v^2 + \sigma_a^2 \mathbf{h}_{R,v}^H \mathbf{\Theta} \mathbf{D}_a \mathbf{\Theta}^H \mathbf{h}_{R,v}}. \quad (16)$$

The achievable ergodic rate at VU v is then $R_v = \mu B \log_2(1 + \Gamma_v)$.

Let $\mathbf{x}_D[k] \in \mathbb{C}^{M \times 1}$ denote the backhaul signal transmitted by the IAB donor. The received signal at IAB node i is

$$y_{\text{BH},i}[k] = \mathbf{h}_{D,i}^H \mathbf{x}_D[k] + z_{\text{BH}}[k], \quad (17)$$

where $\mathbf{h}_{D,i} \in \mathbb{C}^{M \times 1}$ is the donor-IAB channel (LoS-dominant), and $z_{\text{BH}}[k] \sim \mathcal{CN}(0, \sigma_{\text{BH}}^2)$ is adaptive white gaussian noise (AWGN).

The SNR for the backhaul link at IAB node i is

$$\Gamma_{\text{BH},i} = \frac{\|\mathbf{h}_{D,i}^H \mathbf{W}_{\text{BH}}\|^2}{\sigma_{\text{BH}}^2}, \quad (18)$$

where $\mathbf{W}_{\text{BH}} \in \mathbb{C}^{M \times V}$ is the beamforming matrix used by the donor for backhaul transmission.

The achievable backhaul rate between the donor and IAB node i is $R_{\text{BH},i} = (1 - \alpha) B \log_2(1 + \Gamma_{\text{BH},i})$. The effective backhaul rate allocated to VU v is $R_{\text{BH},v} = \frac{R_v}{\sum_{v'=1}^V R_{v'}} R_{\text{BH},i}$, and the resulting end-to-end communication rate is $R_{\text{E2E},v} = \min(R_v, R_{\text{BH},v})$.

The total HRIS power consumption is modeled as

$$P_{\text{HRIS}} = \sum_{n \in \mathcal{N}} \left[d_n (P_{\text{ps}} + P_{\text{bias}} + \zeta^{-1} |\alpha_n|^2 (P_{\text{in},n} + \sigma_a^2)) + (1 - d_n) P_{\text{ps}} \right]. \quad (19)$$

where N_p is the number of passive elements, $\mathcal{N}_a$ the active-element set, P_{ps} is the phase-shifter power, P_{bias} the DC bias of each amplifier, ζ the amplifier efficiency, $P_{\text{in},n}$ the incident RF power at element n , and α_n is the amplification gain.

Finally, the network EE is

$$\eta_{\text{EE}} = \frac{\sum_{v=1}^V R_{\text{E2E},v}}{P_{\text{TX}} + P_{\text{HRIS}} + P_{\text{ckt}}}. \quad (20)$$

Symbol P_{ckt} is the circuit power decomposition at radio nodes and is defined as

$$P_{\text{ckt}} = P_{\text{BB}} + N_{\text{tx}} P_{\text{DAC}} + N_{\text{rx}} P_{\text{ADC}} + M P_{\text{RF}} + P_{\text{LO}} + P_{\text{ctrl}}, \quad (21)$$

where P_{BB} is the baseband power, $P_{\text{DAC}}/P_{\text{ADC}}$ are the data-converter powers, P_{RF} covers power of per-chain mixers/filters/switches, P_{LO} is the power of synthesizer/clocking, and P_{ctrl} includes the power of HRIS driver/controller overhead.

III. EE OPTIMIZATION PROBLEM

We aim to jointly optimize the transmit beamforming vector $\mathbf{w}$ at the IAB node, the HRIS reflection amplitudes $\{\mu\} \in \mu_{r,n} \forall r, n$, the phase shift matrix $\{\Theta\} \in \Theta_r \forall r$, and the element selection vector $\mathbf{s}$ to maximize the overall system EE.

The EE maximization problem is formulated as:

$$\max_{\mathbf{w}, \Theta, \{\mu\}, \mathbf{s}} \eta_{EE} \quad (22a)$$

$$\text{s.t. } R_v \leq R_{BH,i}, \quad \forall v \in \mathcal{V}, i \in \mathcal{I}, \quad (22b)$$

$$\Gamma_v \geq \gamma_{\min}, \quad \forall v \in \mathcal{V}, \quad (22c)$$

$$\text{tr}(\mathbf{W}_1 \mathbf{W}_1^H) + \text{tr}(\mathbf{W}_2 \mathbf{W}_2^H) \leq P_{\text{IAB}}^{\max}, \quad (22d)$$

$$\mu_{r,n} = \begin{cases} \in [1, \mu_{\max}], & \text{if } s_{r,n} = 2 \\ 1, & \text{if } s_{r,n} = 1 \\ 0, & \text{if } s_{r,n} = 0 \end{cases}, \quad (22e)$$

$$P_{\text{HRIS}} \leq P_{\text{HRIS}}^{\max}, \quad (22f)$$

$$s_{r,n} \in \{0, 1, 2\}, \quad \forall n, \quad (22g)$$

$$\sum_{n=1}^N \mathbb{I}(s_{r,n} = 0) = N_o, \quad (22h)$$

$$\sum_{n=1}^N \mathbb{I}(s_{r,n} = 1) + \sum_{n=1}^N \mathbb{I}(s_{r,n} = 2) = N - N_o, \quad (22i)$$

where constraint (22b) ensures that each VU's rate does not exceed its allocated backhaul capacity, (22c) guarantees a minimum SINR, (22d) limits the transmit power of IAB nodes, (22e) bounds the gain of active HRIS elements, (22f) restricts the total HRIS power consumption, (22g) enforces discrete mode selection for each element, (22h) sets the number of Off-mode elements to N_o , and (22i) ensures that the remaining $N - N_o$ elements are sparsely and adaptively assigned to active and passive modes.

IV. PROPOSED SOLUTION

The EE maximization problem in (29a) is high-dimensional and non-convex due to the fractional objective and coupling among the IAB beamforming vector w , HRIS matrix Θ , element selection, and transmit power P_{TX} . Conventional optimization struggles with such nonlinearity and the dynamic vehicular environment. Therefore, we adopt a learning-based solution using the DDPG algorithm, which efficiently handles continuous action spaces and learns optimal policies through environment interaction.

A. State Space Formulation

The DDPG agent observes the current network state s_t to guide the decision-making. The state includes all relevant channel state information affecting EE:

$$s_t = \{\mathbf{h}_{i,v,t}, \mathbf{H}_{i,R,t}, \mathbf{h}_{R,v,t}, \mathbf{h}_{D,i,t}\}, \quad \forall v \in \mathcal{V}, i \in \mathcal{I}. \quad (23)$$

The channel state $\mathbf{h}_{i,v,t}$, $\mathbf{H}_{i,R,t}$, $\mathbf{h}_{R,v,t}$, $\mathbf{h}_{D,i,t}$ is time-indexed so that the state sequence $s_1, s_2, \dots$ and the effective topology evolve across steps. A one-dimensional road layout yields stronger temporal correlation in this evolution than a

Algorithm 1 DDPG with hybrid MLP-LLM Actor

- 1: Randomly initialize critic network $Q(s, a|\theta^Q)$ and Hybrid actor $\mu_{\text{Hybrid}}(s, p|\theta^\mu)$.
- 2: Initialize target networks Q' and μ'_{Hybrid} with weights $\theta^{Q'} \leftarrow \theta^Q$, $\theta^{\mu'} \leftarrow \theta^\mu$.
- 3: Initialize replay buffer R .
- 4: **for** episode = 1, M **do**
- 5: Initialize a random process $\mathcal{N}$ for action exploration.
- 6: Receive initial observation state s_1 .
- 7: **for** t = 1, T **do**
- 8: Generate text prompt p_t from numerical state s_t .
- 9: Select action $a_t = \mu_{\text{Hybrid}}(s_t, p_t|\theta^\mu) + \mathcal{N}_t$.
- 10: Execute a_t , observe r_t and s_{t+1} .
- 11: Generate next prompt p_{t+1} from s_{t+1} .
- 12: Store transition $(s_t, p_t, a_t, r_t, s_{t+1}, p_{t+1})$ in R .
- 13: Sample a random minibatch of N transitions $(s_i, p_i, a_i, r_i, s_{i+1}, p_{i+1})$ from R .
- 14: Set y_i .
- 15: Update critic by minimizing the loss $L(\theta^Q)$.
- 16: Update the actor policy using the sampled policy gradient (24).
- 17: Update the target networks using soft updates.

$$\theta^{Q'} \leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'}$$

$$\theta^{\mu'} \leftarrow \tau \theta^\mu + (1 - \tau) \theta^{\mu'}$$
- 18: **end for**
- 19: **end for**

grid where vehicles change direction randomly at intersections; Section V describes the evaluation setup.

B. Action Space Design

The action vector a_t contains the controllable parameters related to the EE objective: $a_t = \{\mathbf{w}, \Theta, \mu, \mathbf{s}\}$. Here, Θ is defined by the phase and amplitude coefficients of each HRIS element, determined by the mode selection vector $\mathbf{s} = [s_{r,1}, \dots, s_{r,N_r}]^T$.

C. DDPG Framework Overview

The DDPG agent consists of two main components: an actor network $\mu(s|\theta^\mu)$ that maps states to deterministic actions, and a critic network $Q(s, a|\theta^Q)$ that estimates the long-term expected return (the Q-value) for a given state-action pair. DDPG employs target networks for both the actor (μ') and the critic (Q') to ensure stable learning. The critic network is updated by minimizing the loss function: $L(\theta^Q) = \mathbb{E}_{s_i, a_i, r_i, s_{i+1}} [(y_i - Q(s_i, a_i|\theta^Q))^2]$, where $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\theta^{Q'})$ is the target value. The actor network is then updated using the sampled policy gradient to produce actions that maximize the critic's estimated Q-value:

$$\nabla_{\theta^\mu} J \approx \frac{1}{N} \sum_i \nabla_a Q(s, a|\theta^Q)|_{s=s_i, a=\mu(s_i)} \nabla_{\theta^\mu} \mu(s|\theta^\mu)|_{s_i}, \quad (24)$$

D. Reward Function

The goal of the agent is to maximize the overall network EE. Therefore, the reward function r_t at each time step t is defined directly as the objective function in (42), i.e., $r_t = \eta_{EE}$. By maximizing the cumulative sum of this reward, the agent learns a policy that enhances the system's EE.

E. Actor Architecture Variants

As shown in Algorithm 1, the proposed MA-DRL employs parallel MLP and LLM actors to fuse numerical precision with semantic reasoning. An MLP branch encodes the concatenated state vector s_t (including RIS phases and DAM beamforming parameters) through two Rectified Linear Unit-activated hidden layers of 128 neurons each, producing numerical embeddings. In parallel, a pre-trained DistilBERT model processes descriptive prompts about system status to extract semantic embeddings. These representations are concatenated and passed through fully connected layers with tanh activation to generate continuous actions, effectively integrating low-level numerical features with high-level contextual understanding for improved decision-making.

The LLM branch contributes three capabilities that a purely numerical MLP lacks. The self-attention layers of DistilBERT model long-range dependencies in the state sequence; when the prompt encodes recent channel and load history, the attention mechanism correlates events separated by many time steps, enabling proactive HRIS and power reallocation before the bottleneck fully manifests. The tokenized prompt abstracts low-level channel coefficients into semantic concepts, such as access link degrading or backhaul nearing congestion, so that the policy generalizes across episodes with similar high-level dynamics even when channel realizations differ. Because the state evolves step-wise as in Section II and the evaluation uses a one-dimensional layout, the state sequence exhibits temporal correlation that the LLM can learn while the MLP must infer such structure implicitly from raw s_t , being less sample-efficient. The complementary MLP branch retains fine-grained numerical precision for phase shifts, amplification gains, and beamforming weights. Together, the two branches exploit the temporal structure in the state sequence while preserving numerical accuracy.

F. Training Algorithm

The end-to-end training alternates between environment interaction and policy updates, where the agent observes the channel state s_t , selects an action a_t via the actor network with exploration noise, applies it to the environment to obtain the reward r_t and next state s_{t+1} , stores transitions in a replay buffer, and updates the actor-critic and target networks through mini-batch learning, enabling convergence to an energy-efficient control policy under dynamic channels and system constraints.

V. NUMERICAL RESULTS AND DISCUSSION

This section evaluates the proposed MA-DRL framework for EE optimization in sparse HRIS-assisted IAB networks via extensive simulations, benchmarking it against MLP-only and

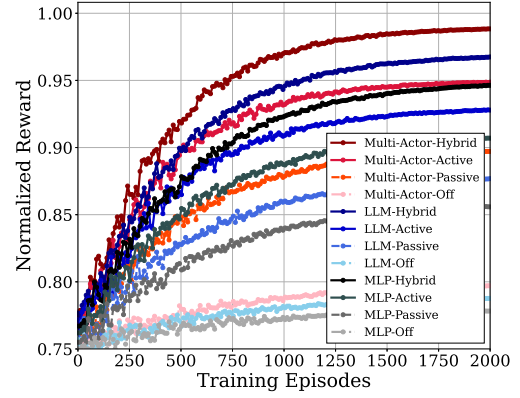


Fig. 2: Convergence of EE over training episodes.

LLM-only DRL baselines under diverse network conditions. Table I summarizes the system parameters and learning hyper-parameters used in our simulations. The setup includes an IAB donor, a single IAB node, and one HRIS serving multiple VUs. The evaluation uses a one-dimensional road segment of length 100 m with the IAB node at 0 m, HRIS at 30 m, and VU positions in $[50, 80]$ m along the segment, with the donor at 0 m; V varies from 2 to 12 as in Table I. At each DRL step the channel state is updated so that the state s_t and effective topology evolve over the episode. In all figures, we consider the following baseline schemes: (i) MLP-based DRL, (ii) LLM-based DRL, (iii) passive RIS, (iv) active RIS, and (v) No-RIS.

TABLE I: Key System Parameters and Learning Hyper-parameters

Parameter	Value
Carrier frequency	28 GHz (mmWave)
Bandwidth (B)	200 MHz
Bandwidth split factor (α)	0.5
IAB node antennas (M)	16
HRIS elements (N_r)	[4–64]
Vehicular users (V)	[2–12]
Max IAB transmit power ($P_{\max}$)	10–40 dBm
Training episodes	3000
Actor hidden layers	[256, 128] (MLP branch)
LLM embedding dimension	384 (DistilBERT-based)
Optimizer	Adam ($\eta_{\text{actor}} = 10^{-4}$, $\eta_{\text{critic}} = 10^{-3}$)

Fig. 2 depicts the convergence behavior over 2000 training episodes for three DRL algorithms (MLP, LLM, and MA) under for RIS modes: hybrid, active, passive, and off. For all RIS cases, the results show rapid early learning and smooth convergence, with the MA-DRL improving final normalized reward by approximately 2.15% and 4.40% over the LLM and MLP-based DRLs, respectively. This performance gain arises from the synergy between semantic understanding and precise numerical optimization, allowing better adaptation to channel variations. Furthermore, the MA-HRIS consistently outperforms the MA-Active and MA-Passive cases, yielding 4.21% and 10.0% higher final normalized rewards at convergence, respectively.

Fig. 3a shows the effect of the number of RIS elements on EE, comparing hybrid, active, passive, and off-mode configurations across the three DRL algorithms. EE increases as the number of RIS elements grows from 0 to 64 due to the additional spatial degrees of freedom that enable constructive signal combining and interference suppression. The sparse

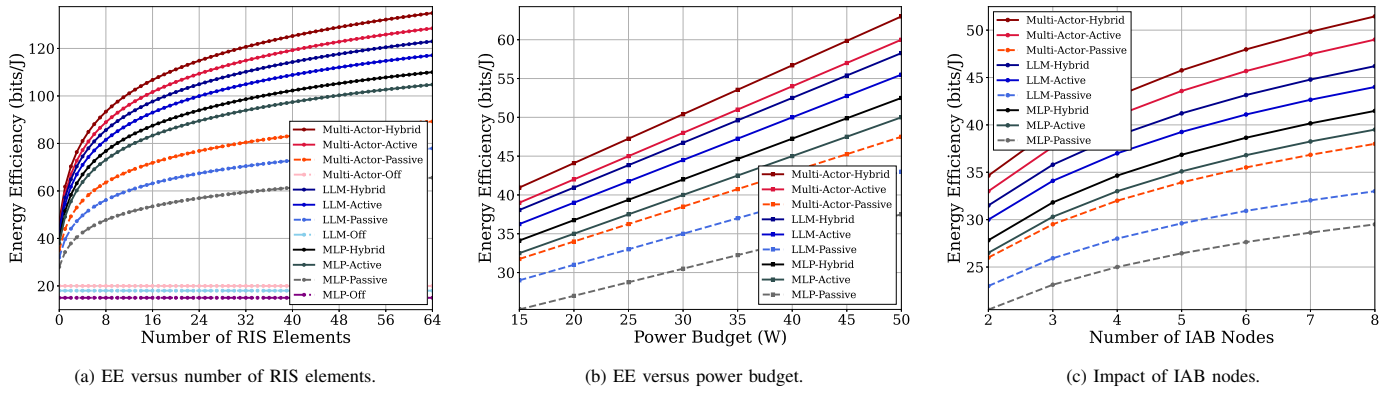


Fig. 3: Energy-efficiency (EE) performance under varying system parameters.

HRIS consistently achieves superior EE because it adaptively combines active amplification, passive reflection, and off-mode deactivation of redundant elements. This selective operation allows HRIS to exploit high beamforming gains when needed while reducing circuit power through element sparsification. Consequently, the MA-HRIS curve remains above the MA-Active curve across the entire 0–64 range, yielding 2.7% and 27% higher EE compared to the MA-Active and MA-Passive cases, respectively.

Fig. 3b illustrates the EE performance as a function of the transmit power budget for the three actor architectures under active, passive, and hybrid RIS modes. EE increases with the available power budget since higher transmit power enhances signal strength and enables more flexible beam and resource allocation. The proposed MA-DRL consistently outperforms the single-actor DRL baselines by dynamically learning the nonlinear trade-off between transmit power consumption and achievable rate, thereby maximizing the rate-to-power ratio under varying channel conditions. In contrast to purely active or passive surfaces, the MA-HRIS achieves additional gains through adaptive mode switching, where only a subset of elements operate in active mode while others remain passive or off, effectively controlling circuit power without compromising array gain. This adaptive hybrid control yields 4.5% and 28% higher EE compared to the MA-Active and MA-Passive configurations, respectively, across the 15–50 W power range.

Fig. 3c presents the EE performance as a function of the number of IAB nodes for the three DRL algorithms. As the number of IAB nodes increases, EE improves through spatial reuse of backhaul spectrum and traffic load distribution, which reduce per-node transmit power and access link congestion. RIS enhances access links by enabling coherent combining over direct and reflected paths, alleviating bottlenecks and improving end-to-end efficiency. The backhaul remains direct LoS between donor and IAB nodes. The proposed MA-DRL framework consistently outperforms the single-actor DRL baselines by learning the joint dynamics of multi-hop routing, interference coordination, and access-backhaul coupling. The LLM actor captures long-range dependencies in topology evolution and backhaul congestion, while the MLP actor provides precise power and beam control at each node. Together they enable context-aware multi-hop optimization. As

the IAB topology becomes denser, the MA-DRL maintains stable convergence and adaptive energy allocation, demonstrating superior scalability in complex hierarchical IAB networks.

VI. CONCLUSIONS

This paper investigated an EE framework for sparse HRIS-assisted IAB networks for next-generation IoT and vehicular deployments. We formulated a joint optimization problem to maximize EE of the network by controlling IAB beamforming, sparse HRIS configuration, and power allocation under realistic hardware constraints. We developed a DDPG-based framework with an MA-DRL algorithm, benchmarked against MLP-only and LLM-only baselines. Simulation results showed that the proposed MA-DRL achieves 17.4% and 42.1% higher EE than the LLM-based and MLP-based approaches, respectively.

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